

DUVAN, Russian Federation

WMO: 285370

Lat: 55.716N

Lon: 57.899E

Elev: 1109

StdP: 14.12

Time Zone: 5.00 (E05)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.4	-16.2	-25.8	1.4	-20.9	-20.8	1.8	-15.7	29.7	28.3	26.6	23.5	3.8	200	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	83.7	66.5	80.4	64.8	77.2	63.3	68.7	80.2	67.0	78.0	65.0	74.6	6.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	93.9	74.6	62.7	88.7	72.4	61.1	83.8	70.1	33.5	80.8	32.1	78.1	30.5	74.8	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.2	21.7	16.5	-29.0	88.2	5.9	2.2	-33.2	89.7	-36.6	91.0	-39.9	92.2	-44.2	93.8		
(5)			-29.1	71.0	5.9	2.9	-33.4	73.1	-36.8	74.8	-40.1	76.4	-44.4	78.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	9.6	10.9	22.7	37.6	51.4	60.4	64.3	59.3	49.0	38.3	19.7	10.1	
	DBStd	22.39	12.21	13.25	8.68	10.40	9.18	7.51	6.60	7.34	8.21	8.59	12.48	13.69	
	HDD50	6337	1251	1096	847	391	98	9	0	5	113	379	910	1238	
	HDD65	10653	1716	1516	1312	822	427	176	98	212	482	829	1360	1703	
	CDD50	1315	0	0	0	19	142	320	442	293	84	15	0	0	
	CDD65	157	0	0	0	0	6	37	76	35	3	0	0	0	
	CDH74	1462	0	0	0	3	104	375	662	285	33	0	0	0	
	CDH80	323	0	0	0	0	6	87	142	79	9	0	0	0	
Wind	WSAvg	5.9	5.8	6.5	6.5	6.3	7.2	5.7	4.6	4.5	5.2	6.2	5.5	6.4	
Precipitation	PrecAvg	21.6	1.1	0.9	1.0	1.1	2.0	2.6	2.8	2.8	2.0	2.2	1.6	1.3	
	PrecMax	29.8	2.3	2.0	2.3	2.4	4.4	7.5	7.2	5.3	4.2	4.3	3.6	3.6	
	PrecMin	14.3	0.0	0.1	0.0	0.1	0.2	0.6	0.4	0.7	0.4	0.3	0.2	0.4	
	PrecStd	4.5	0.6	0.5	0.6	0.6	1.1	1.5	1.5	1.1	1.0	1.1	0.9	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	35.1	43.7	71.9	80.5	86.9	87.3	86.7	78.0	66.9	42.0	33.7	
		MCWB	31.2	32.4	37.5	52.0	56.8	67.0	68.3	68.2	59.6	53.2	38.4	32.1	
	2%	DB	31.0	32.5	38.9	65.0	77.2	82.2	83.7	81.6	71.5	59.6	38.3	32.0	
		MCWB	29.4	30.8	34.3	49.1	57.5	64.8	67.2	67.2	56.8	50.3	36.3	31.1	
	5%	DB	28.4	30.3	36.2	60.1	73.0	78.1	81.0	76.1	66.4	54.7	35.8	30.3	
		MCWB	27.1	28.9	32.6	47.8	56.6	63.6	66.7	64.1	55.4	48.2	33.7	29.3	
	10%	DB	26.2	28.3	34.0	53.6	68.4	74.3	77.7	71.0	61.8	50.4	34.0	28.0	
		MCWB	25.1	27.0	31.3	44.4	54.2	62.0	65.2	61.8	53.2	45.9	32.5	26.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.9	32.4	39.0	53.6	61.9	69.4	70.8	70.3	63.1	54.6	39.4	32.8
			MCDB	32.7	34.2	42.2	68.9	73.6	81.0	82.6	83.1	77.1	61.5	40.3	33.6
2%		WB	29.5	31.1	35.1	50.7	59.6	67.2	69.2	67.4	58.6	52.2	36.4	31.2	
		MCDB	30.6	32.4	37.7	62.9	71.9	79.4	80.6	79.4	67.2	59.1	37.8	31.8	
5%		WB	27.3	29.2	33.3	48.1	57.5	64.7	67.7	64.9	56.2	49.1	34.5	29.5	
		MCDB	28.4	30.3	35.7	59.4	70.2	75.2	78.5	74.5	63.8	53.5	35.5	30.2	
10%		WB	25.1	27.0	31.6	44.9	55.4	62.8	66.1	62.7	54.0	46.3	32.7	27.0	
		MCDB	26.2	28.4	33.7	53.9	66.2	72.1	76.0	69.8	60.8	50.2	33.7	27.9	
Mean Daily Temperature Range	5% DB	MDBR	11.0	12.5	13.3	16.5	18.4	18.2	18.1	15.4	14.6	9.8	9.8	10.8	
		MCDBR	9.3	9.7	13.1	25.1	26.5	24.6	23.2	22.4	21.9	18.1	10.8	11.8	
		MCWBR	8.9	8.7	9.4	12.8	11.9	11.1	9.8	9.7	10.5	10.4	9.2	11.2	
	5% WB	MCDBR	9.9	8.8	10.8	24.5	24.1	22.3	21.0	19.7	19.2	16.7	8.7	11.7	
		MCWBR	9.6	8.4	8.3	13.4	12.1	11.2	9.8	9.3	10.4	10.9	8.7	11.4	
Clear-Sky Solar Irradiance	taub	0.290	0.314	0.369	0.407	0.401	0.393	0.415	0.432	0.380	0.351	0.312	0.288		
	taud	2.037	2.031	1.942	2.078	2.241	2.330	2.270	2.194	2.303	2.262	2.140	2.001		
	Ebn at Noon	184	229	243	255	266	270	260	246	244	218	176	149		
	Edh at Noon	21	31	44	44	40	37	39	40	31	25	19	17		
All-Sky Solar Radiation	RadAvg	228	535	971	1341	1622	1750	1728	1342	849	383	183	135		
	RadStd	31	66	94	156	166	190	171	148	77	84	37	29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page